Six-theme soft HMM

Training report for the humpback song model in song_soft_hmm_t6.dot. Baum–Welch fits the slot Gaussians, the duration exits, the theme-end branches, and the start distribution. The motif scores are the observations. This report does not decode a state path.

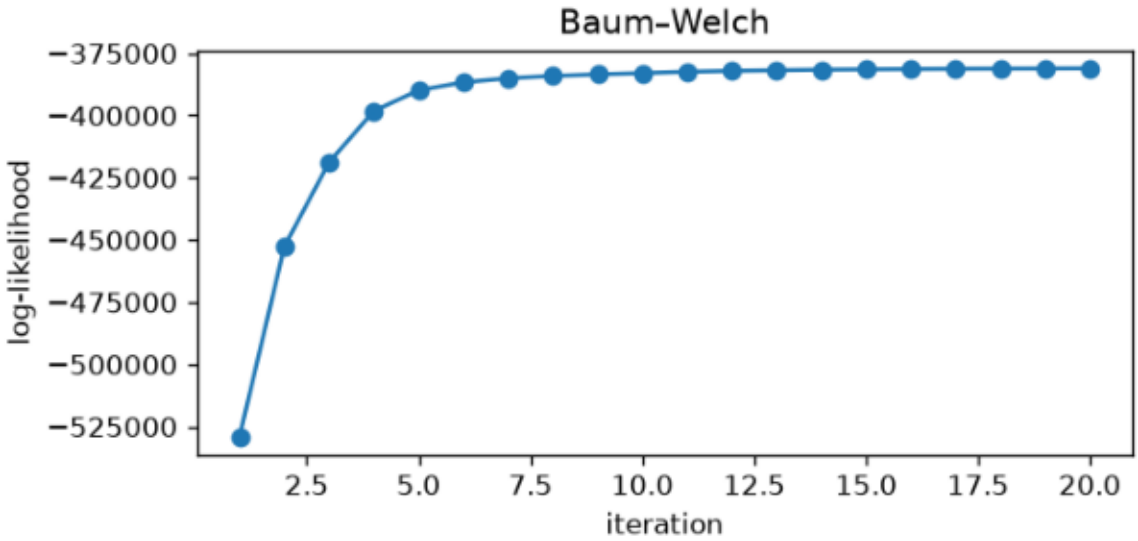
Recording and features

File: SOCAL_H_76_230426_193500_df100.x.x_10kHz.wav. 10,000 Hz, mono, 36,000 seconds. The file was upsampled from 2 kHz, so GPL was limited to 10–1000 Hz. GPL wrote 480 whitened windows of 75 s under analysis/HMM/work/gpl_193500/.

Each window was summed over frequency and averaged to 10 Hz (360,000 samples). panMP kept the 12 strongest motif peaks, all 1 second long. Each second of the recording is the mean z-normalized correlation with each prototype, then each score is z-scored across the 10 hours. The motif CSV cuts the 10 hours into 53 songs and initializes the Gaussians. It is not the observation.

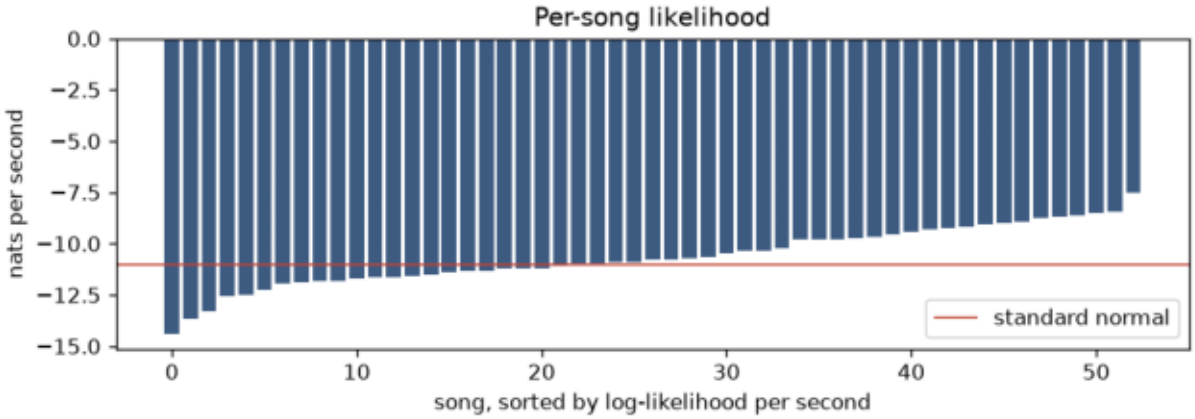
Sequence log-likelihood

Baum–Welch shared one parameter set across the 53 songs and ran for 20 iterations. The E-step log-likelihood moved from -528,963.4 to -381,054.1. The saved parameters score the 36,000 seconds at -381,018.0, which is -10.584 nats per second. Twelve independent standard normals, the distribution the z-score produces when every slot mean stays at zero, score -11.027 nats per second. The fit is +0.443 nats per second relative to that baseline. The initialization scored -528,963.4.



Where the likelihood comes from

Per song, the rate averages -10.639 nats per second (median -10.756, lowest -14.409, highest -7.549). A song far below the others is one the shared parameters explain poorly.



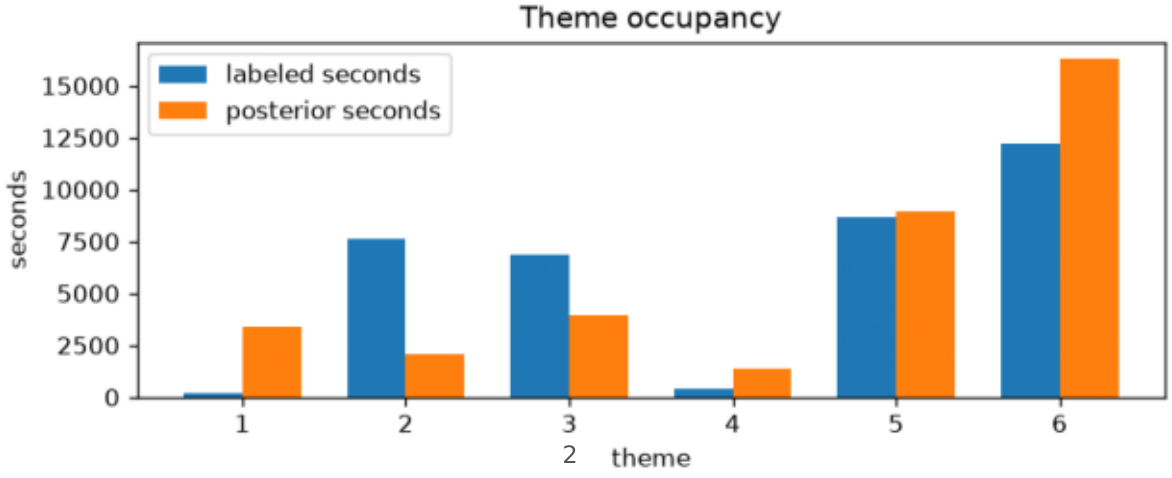
Complete-data split

The posterior-weighted complete-data log-likelihood on the saved parameters splits into emission, transition, and start. Emission -358,251.8. Transitions -23,271.3. Start -67.9. The last training E-step, on the parameters before the final update, was -381,624.0.

Posterior mass on the labeled theme

At each second the forward–backward pass gives a probability for each theme. The number below is the average of the probability on the theme the CSV wrote for that second. Across all seconds it is 40.9 percent. It is a property of the fitted distribution, not a chosen path.

Theme	labeled s	posterior s	mass on label
1	216	3376	8.3%
2	7623	2042	6.3%
3	6896	3975	17.5%
4	376	1394	5.8%
5	8671	8927	40.7%
6	12218	16285	77.6%



The fitted parameters

The drawing has 28 phrase slots: 3, 3, 6, 3, 5, and 8 across themes 1 through 6. Each slot expands to five duration states, for 140 states, sharing one diagonal Gaussian over the 12 motif scores. A slot may end after 1–4 seconds and must end at 5. Only the last slot of a theme chooses repeat versus next. After theme 6, next returns to theme 1. A sequence may end in any theme. A song may also open on the first slot of any theme. Slots that share a unit name do not share a Gaussian.

Theme-end probabilities

Repeat is the probability of returning to the first slot of the same theme when a phrase ends. The drawing supplied the starting values. Labeled phrases per song is how many times the CSV repeats that theme's slot chain.

Theme	slots	start repeat	fitted repeat	labeled phrases/song
1	3	0.008	0.728	1.00
2	3	0.984	0.602	40.94
3	6	0.943	0.572	20.60
4	3	0.585	0.435	2.28
5	5	0.922	0.831	31.19
6	8	0.950	0.855	21.83

Fitted probability of opening on themes 1 through 6: 0.102 0.068 0.085 0.136 0.051 0.559

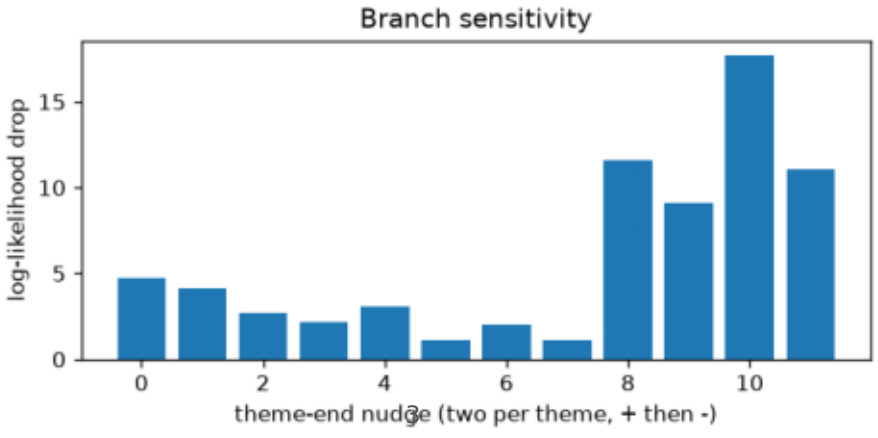
The least-used slot carries 368.1 expected seconds. 0 slots have less than one expected second.

Parameter sensitivity

Each free theme-end probability was nudged by 0.05, and each slot mean was shifted by a tenth of its scale on the first motif score. Nothing was refit. The drop is how far the sequence log-likelihood of all 53 songs fell.

Theme 1: +0.05 drop 4.8, −0.05 drop 4.1.
Theme 2: +0.05 drop 2.7, −0.05 drop 2.1.
Theme 3: +0.05 drop 3.1, −0.05 drop 1.1.
Theme 4: +0.05 drop 2.0, −0.05 drop 1.1.
Theme 5: +0.05 drop 11.6, −0.05 drop 9.1.
Theme 6: +0.05 drop 17.7, −0.05 drop 11.1.

The median branch drop is 3.6. The two largest drops are the +0.05 nudges on theme 6 (17.7) and theme 5 (11.6). Across the 28 slot-mean shifts, the median drop is 5.9 and the largest is 12.8.



The parameter file

song_soft_hmm_t6.npz is the handoff. It holds the slot means and variances, the duration-exit probabilities, the theme-repeat probabilities, the start distribution, the 12 prototypes, and the score mean and standard deviation. A later decoder loads that file. This folder does not choose a state path.

Where the pieces are

Parameter file: analysis/HMM/song_soft_hmm_t6.npz.
Topology: analysis/HMM/topology.py, matching song_soft_hmm_t6.dot.
Baum–Welch: analysis/HMM/hmm.py.
GPL collapse, panMP prototypes, and scoring: analysis/HMM/observations.py.
Training entry: analysis/HMM/run_train.py.
Notebook: analysis/HMM/train_song_soft_hmm_t6.ipynb.
Metrics: analysis/HMM/work/train_report.json.
This report: analysis/HMM/song_soft_hmm_t6_report.pdf.

Retraining is python -m analysis.HMM.run_train from the repository root. It reuses the GPL windows and the motif-score file when they are already on disk, then overwrites the parameter file. python -c "from analysis.HMM.run_train import refresh_metrics; refresh_metrics()" recomputes the metrics for the saved parameters and does not run Baum–Welch again.